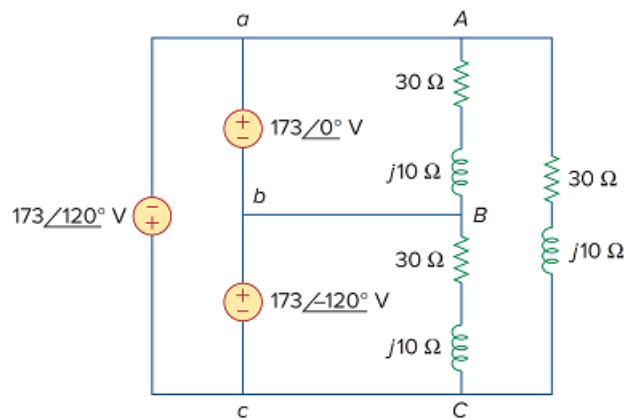


Problem

For the Δ - Δ circuit of Fig. 12.50, calculate the phase and line currents.

Figure 12.50



Step-by-step solution

Step 1 of 2

$$Z_{\Delta} = 30 + j10 = 31.62 \angle 18.43^{\circ}$$

The phase currents are,

$$I_{AB} = \frac{V_{ab}}{Z_{\Delta}} = \frac{173 \angle 0^{\circ}}{31.62 \angle 18.43^{\circ}} = 5.47 \angle -18.43^{\circ} \text{ A}$$

$$I_{BC} = I_{AB} \angle -120^{\circ} = 5.47 \angle -138.43^{\circ} \text{ A}$$

$$I_{CA} = I_{AB} \angle 120^{\circ} = 5.47 \angle 101.57^{\circ} \text{ A}$$

Step 2 of 2

The line currents are,

$$I_a = I_{AB} - I_{CA} = I_{AB} \sqrt{3} \angle -30^{\circ}$$

$$= 5.47 \sqrt{3} \angle -48.43^{\circ} = 9.474 \angle -48.43^{\circ} \text{ A}$$

$$I_b = I_a \angle -120^{\circ} = 9.474 \angle -168.43^{\circ} \text{ A}$$

$$I_c = I_a \angle 120^{\circ} = 9.474 \angle 71.57^{\circ} \text{ A}$$